Minimal Surfaces and the Polyakov Action

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1. Polyakov action and classical equivalence

One starts from the Nambu–Goto action for a bosonic string in a flat D-dimensional Euclidean target space:

$$S_{NG} = -\frac{T}{2}\int d^2\sigma\,\sqrt{\det \partial_a X^\mu \partial_b X_\mu}, \qquad \mu = 1,\dots,D. \tag{1}$$

The Polyakov action is

$$S_P = -\frac{T}{2}\int d^2\sigma\,\sqrt{h}\,h^{ab}\,\partial_a X^\mu \partial_b X_\mu, \qquad \mu = 1,\dots,D, \tag{2}$$

with worldsheet metric h_{ab} and tension T.

To see why varying only h_{ab} reproduces the Nambu–Goto action, write temporarily $\partial_a X^\mu \partial_b X_\mu$ as M_{ab}, so that the problem is to maximize

$$\sqrt{h}\,h^{ab}\,M_{ab},$$

whose maximum is easily seen to be $\sqrt{\det M_{ab}}$ by diagonalizing M_{ab}.

Varying h_{ab} gives the stress-energy tensor constraint $T_{ab} = 0$, and eliminating h_{ab} via its equation of motion reproduces the Nambu–Goto (area) action. Thus, classically, S_P describes minimal surfaces.

2. Conformal gauge

Using 2D diffeomorphism and Weyl invariance we choose the **conformal gauge** $h_{ab} = e^{2\phi}\delta_{ab}$. In Euclidean worldsheet coordinates (τ,σ) (or complex coordinates $z = \tau + i\sigma$, $\bar z = \tau - i\sigma$) the action becomes

$$S = \frac{1}{2}\int d^2z\;\partial X^\mu \bar\partial X_\mu, \qquad \text{with } \partial = \partial_z,\ \bar\partial = \partial_{\bar z}, \tag{3}$$

where we have absorbed T into the field normalization for simplicity (or set $T = 1/\pi$). The equations of motion are $\partial\bar\partial X^\mu = 0$, so each X^μ is a **harmonic function** on the worldsheet.

3. Virasoro constraints

Gauge fixing leaves residual constraints $T_{ab} = 0$. In complex coordinates they reduce to

$$T_{zz} \equiv (\partial X)^2 = 0, \tag{4}$$

$$T_{\bar z\bar z} \equiv (\bar\partial X)^2 = 0. \tag{5}$$

These are the **Virasoro constraints**, enforcing that the worldsheet coordinates are isothermal (conformal) on the induced metric.

4. Solving the constraints in $\mathbb{R}^3$

For a minimal surface in $\mathbb{R}^3$, define the holomorphic differential

$$\phi_i(z) = \partial X_i(z, \bar{z}), \qquad i = 1, 2, 3.$$

Each ϕ_i is holomorphic because $\partial\bar{\partial} X_i = 0$. Equation (4) becomes the complex constraint

$$\phi_1^2 + \phi_2^2 + \phi_3^2 = 0. \tag{6}$$

This is a single complex equation for three holomorphic functions.

5. Weierstrass–Enneper parametrisation

Recall the primary school equalities $\sin^2\theta + \cos^2\theta = 1$ and $\sin\theta = \frac{2\tan(\theta/2)}{\tan(\theta/2)^2+1}$, $\cos\theta = \frac{\tan(\theta/2)^2-1}{\tan(\theta/2)^2+1}$. So we can reparameterize the solutions of (6) by two holomorphic functions $f(z)$ and $g(z)$ (with f holomorphic, g meromorphic, and f having a zero of order $2m$ wherever g has a pole of order m). The components are

$$\boxed{\phi_1 = \frac{1}{2} f\,(1 - g^2), \qquad \phi_2 = \frac{i}{2} f\,(1 + g^2), \qquad \phi_3 = f\,g.} \tag{7}$$

Indeed,

$$\phi_1^2 + \phi_2^2 + \phi_3^2 = \frac{f^2}{4}\left[(1-g^2)^2 - (1+g^2)^2\right] + f^2 g^2 = -f^2 g^2 + f^2 g^2 = 0.$$

The embedding $X_i(z, \bar{z})$ is recovered by integration (taking the real part for a real surface):

$$X_i(z, \bar{z}) = \mathrm{Re} \int \phi_i(z)\, dz. \tag{8}$$

The integral is path independent (up to possible periods) because ϕ_i is holomorphic. The resulting surface automatically satisfies both the harmonic equation and the Virasoro constraints, hence is a **minimal surface**.

6. Bernstein's theorem

Theorem 1. *Let $u(x, y)$ be a C^2 solution of the minimal surface equation on the whole plane $\mathbb{R}^2$. Then $u(x, y)$ is a linear function, i.e. its graph is a plane.*

Proof. A minimal graph $X(x, y) = (x, y, u(x, y))$ can be re-parametrised isothermally. The Weierstrass–Enneper representation then provides two holomorphic data functions $f(z)$ and $g(z)$ (with $z = x + iy$) that generate the surface.

For a graph over the entire (x, y)-plane, the Gauss map is one-to-one onto a subset of the upper hemisphere (or lower, depending on orientation). Stereographic projection of the unit normal gives the holomorphic function $g(z)$, which maps the plane $\mathbb{C}$ into the unit disk $\mathbb{D}$ (or its complement).

More precisely, the gradient of u is bounded away from infinity because the normal never becomes horizontal—otherwise the graph would have a vertical tangent. Thus the image of g omits an open set of the Riemann sphere.

Since $g(z)$ is meromorphic and its image omits at least two points (it avoids a hemisphere), Picard's little theorem forces $g(z)$ to be constant. (One can also argue more simply: g is a bounded entire function, hence constant by Liouville's theorem.)

With $g(z) \equiv c$, the Weierstrass–Enneper differential becomes

$$\phi_1 = \frac{f}{2}(1 - c^2), \quad \phi_2 = \frac{if}{2}(1 + c^2), \quad \phi_3 = fc.$$

Integrating and taking real parts gives

$$X_1 + iX_2 = \frac{1}{2}(1 - c^2) \int f(z)\, dz + \frac{i}{2}(1 + c^2) \int f(z)\, dz + \text{const.}$$

But for the graph parametrisation we need $X_1 = x$, $X_2 = y$ (up to a complex reparametrisation). This forces $f(z)$ to be a real constant (or to produce a linear function), which in turn makes $u(x, y)$ linear.

A standard form of the proof: using the Weierstrass data, one shows that if the surface is an entire graph, then $|g| < 1$ everywhere (the normal points consistently into the upper half-space). g is then a bounded holomorphic function, hence constant. A constant g gives ϕ_3/ϕ_1 and ϕ_3/ϕ_2 constant, implying the normal is fixed—the surface is a plane. $\square$